Complete C# + .NET Web API Roadmap

Compact SQL Server notebook-style master notes - numbered topics - bulleted subtopics

Coverage: 40 ordered topics and 1,525 supplied subtopics. Topics continue one after another with no chapter-cover gaps. Three compact reference diagrams are included only where they improve understanding.

1. C# and .NET Fundamentals

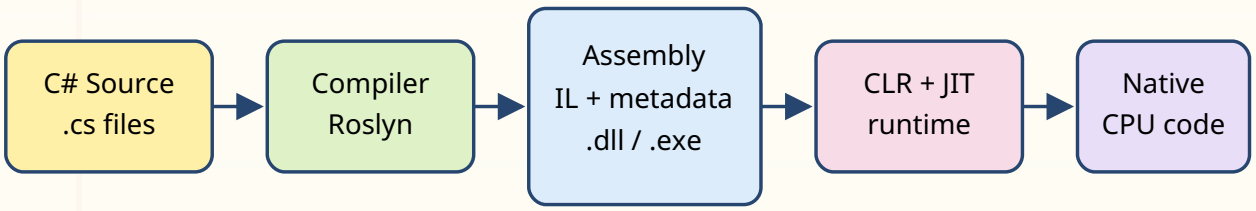
40 subtopics

- C# language
- .NET platform
- .NET SDK
- .NET Runtime
- CLR
- Managed code
- Unmanaged code awareness
- Intermediate Language
- JIT compilation
- Assemblies
- Assembly metadata
- Executables
- Class libraries
- Namespaces
- Using directives
- Solution files
- Project files
- `.csproj`
- Project references
- Package references
- NuGet
- .NET CLI
- `dotnet new`
- `dotnet restore`
- `dotnet build`
- `dotnet run`
- `dotnet test`
- `dotnet clean`
- `dotnet publish`
- Debug configuration
- Release configuration
- Top-level statements
- `Program.cs`
- Compiler errors
- Compiler warnings
- Debugging
- Breakpoints
- Call stack
- Watch variables

- Local variables window

REFERENCE DIAGRAM 1

C# / .NET Compilation and Runtime Map



2. Variables, Types and Nullability

41 subtopics

- Variables
- Local variables
- Fields
- Constants
- Type inference
- `var`
- `object`
- `dynamic`
- `bool`
- `byte`
- `sbyte`
- `short`
- `ushort`
- `int`
- `uint`
- `long`
- `ulong`
- `float`
- `double`
- `decimal`
- `char`
- `string`
- `Guid`
- `DateTime`
- `DateTimeOffset`
- `DateOnly`
- `TimeOnly`
- `TimeSpan`
- Default values
- `default`
- Nullable value types
- `Nullable<T>`
- Nullable reference types
- Nullable context

- Nullable annotations
- Nullable warnings
- Null-forgiving operator
- Required members
- Numeric limits
- Overflow
- Underflow awareness

3. Operators, Conversion and Pattern Matching

38 subtopics

- Arithmetic operators
- Assignment operators
- Comparison operators
- Logical operators
- Bitwise operators
- Shift operators
- Increment
- Decrement
- Conditional operator
- Null-coalescing operator
- Null-coalescing assignment
- Null-conditional operator
- Operator precedence
- Operator associativity
- Implicit conversion
- Explicit conversion
- Casting
- Numeric conversion
- Reference conversion
- `Convert`
- `Parse`
- `TryParse`
- `checked`
- `unchecked`
- `is`
- `as`
- `typeof`
- `nameof`
- `sizeof` awareness
- Constant patterns
- Type patterns
- Property patterns
- Relational patterns
- Logical patterns
- Positional patterns
- List patterns
- Switch expressions
- Pattern matching with null

4. Control Flow and Decision Making

23 subtopics

- `if`

- `else`
- `else if`
- Nested conditions
- `switch`
- Switch statements
- Switch expressions
- `for`
- `foreach`
- `while`
- `do while`
- Nested loops
- `break`
- `continue`
- `return`
- Guard clauses
- Early return
- Conditional expressions
- Loop control
- Infinite loops
- Avoiding deep nesting
- Readable condition design
- Pattern-based control flow

5. Methods and Parameters

33 subtopics

- Method declaration
- Method invocation
- Return types
- `void`
- Parameters
- Arguments
- Value parameters
- Optional parameters
- Named arguments
- Method overloading
- `params`
- `ref`
- `out`
- `in`
- Passing value types
- Passing reference types
- Local functions
- Recursive methods
- Static methods
- Instance methods
- Expression-bodied methods
- Extension methods
- Generic methods
- Method groups
- Pure methods
- Side effects

- Method scope
- Method accessibility
- Method design
- Guard clauses in methods
- Returning tuples
- Returning records
- Async method signatures awareness

6. Strings, Arrays, Indexes and Ranges

33 subtopics

- String creation
- String immutability
- String interpolation
- String concatenation
- String formatting
- Composite formatting
- String comparison
- Ordinal comparison
- Case-insensitive comparison
- Searching
- Splitting
- Joining
- Replacing
- Trimming
- Padding
- Substrings
- `StringBuilder`
- Character operations
- Arrays
- Single-dimensional arrays
- Multidimensional arrays
- Jagged arrays
- Array initialization
- Array iteration
- Array searching
- Array sorting
- Array copying
- `Array` helper methods
- Index operator
- Range operator
- Slicing
- Arrays vs collections
- Array performance basics

7. Value Types, Reference Types and Memory Basics

24 subtopics

- Value types
- Reference types
- Copy semantics
- Reference copying
- Object references
- Passing arguments

- Mutability
- Immutability
- Stack concepts
- Managed heap
- Object allocation
- Object lifetime
- Null references
- Boxing
- Unboxing
- Boxing performance
- Value-type boxing through interfaces
- Reference identity
- Value equality basics
- Heap allocation basics
- Garbage collection basics
- Memory pressure awareness
- Struct allocation basics
- Reference escape awareness

8. Classes, Objects and Object Members

31 subtopics

- Class
- Object
- Instance
- Object creation
- Fields
- Properties
- Methods
- Constructors
- Object state
- Object behavior
- Object identity
- Instance members
- Static members
- Auto-properties
- Backing fields
- Getter
- Setter
- `init`
- `required`
- Object initializers
- Collection initializers
- Anonymous objects
- Partial classes
- Nested classes
- Static classes
- Sealed classes
- Primary constructors
- Expression-bodied members
- Class vs object
- Class vs struct

- Class vs record

9. Constructors and Object Initialization

26 subtopics

- Constructors
- Default constructor
- Parameterless constructor
- Parameterized constructor
- Constructor overloading
- Constructor chaining
- `this()`
- `base()`
- Primary constructors
- Static constructors
- Private constructors
- Protected constructors
- Constructor execution order
- Base constructor execution
- Derived constructor execution
- Field initialization order
- Property initialization
- Readonly initialization
- Object initializer
- Required members
- `init`
- Constructor validation
- Constructor dependency injection
- Factory method vs constructor
- Constructor side effects
- Avoiding heavy constructors

10. Encapsulation, Properties and Access Control

30 subtopics

- Encapsulation
- Data hiding
- Private state
- Public behavior
- Fields
- Properties
- Auto-properties
- Backing fields
- Computed properties
- Read-only properties
- Write restrictions
- Property validation
- Object invariants
- Mutable objects
- Immutable objects
- `public`
- `private`
- `protected`
- `internal`

- `protected internal`
- `private protected`
- Default accessibility
- Type accessibility
- Member accessibility
- Constructor accessibility
- Nested type accessibility
- Assembly visibility
- Least accessibility principle
- Public API surface
- Information hiding

11. Static, Const, Readonly and Immutability

25 subtopics

- Instance members
- Static members
- Static fields
- Static properties
- Static methods
- Static classes
- Static constructors
- `const`
- `readonly`
- `static readonly`
- Compile-time constants
- Runtime constants
- Readonly references
- Immutable objects
- Immutable collections awareness
- Static initialization
- Shared state
- Mutable static state
- Thread safety
- Static helper classes
- Static vs instance
- Static vs singleton
- Const vs readonly
- Readonly vs immutable
- When not to use static state

12. OOP Principles and Relationships

29 subtopics

- Object-Oriented Programming
- Class
- Object
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism
- IS-A relationship
- HAS-A relationship
- USES-A relationship

- Association
- Aggregation
- Composition
- Dependency
- One-to-one relationships
- One-to-many relationships
- Many-to-many relationships
- Object ownership
- Object lifetime
- Composition over inheritance
- Coupling
- Cohesion
- Loose coupling
- High cohesion
- Reusability
- Maintainability
- Domain modeling
- Behavior-rich objects
- Anemic model awareness

13. Inheritance and Polymorphism

33 subtopics

- Base class
- Derived class
- Parent class
- Child class
- Single inheritance
- Multilevel inheritance
- Hierarchical inheritance
- Multiple class inheritance limitation
- Multiple inheritance with interfaces
- `base`
- Protected members
- Inherited properties
- Inherited methods
- Constructor execution order
- Virtual methods
- `virtual`
- `override`
- `new`
- Method overriding
- Method hiding
- Compile-time polymorphism
- Runtime polymorphism
- Method overloading
- Runtime dispatch
- Base references
- Derived objects
- Upcasting
- Downcasting
- Safe casting

- Polymorphic collections
- Sealing inheritance
- Fragile base-class problem
- Composition vs inheritance

14. Abstraction, Abstract Classes and Interfaces

31 subtopics

- Abstraction
- Implementation hiding
- Abstract classes
- Abstract methods
- Abstract properties
- Abstract members
- Concrete members
- Abstract constructors
- Partial implementation
- Template behavior
- Interfaces
- Interface declaration
- Interface implementation
- Multiple interfaces
- Interface inheritance
- Interface methods
- Interface properties
- Interface events
- Explicit interface implementation
- Implicit interface implementation
- Default interface members
- Interface references
- Interface polymorphism
- Interface segregation
- Programming against abstractions
- Interfaces with DI
- Interfaces for testing
- Abstract class vs interface
- Interface vs concrete class
- Abstract vs virtual
- Avoiding unnecessary abstractions

15. Structs, Records, Enums, Tuples and Equality

38 subtopics

- Struct
- Struct constructors
- Struct methods
- Struct properties
- Struct copying
- Mutable structs
- Immutable structs
- Readonly structs
- Record structs
- `ref struct` awareness
- Struct vs class

- Records
- Record classes
- Positional records
- Value equality
- `with` expressions
- Record inheritance
- Records vs classes
- Enums
- Enum parsing
- Enum conversion
- Enum validation
- Flags enums
- Tuples
- Named tuples
- Tuple deconstruction
- Tuples vs DTOs
- Reference equality
- Value equality
- `Equals`
- `ReferenceEquals`
- `GetHashCode`
- `IEquatable<T>`
- `IEqualityComparer<T>`
- `IComparable<T>`
- `IComparer<T>`
- Equality operators
- Hash-code rules

16. Collections

31 subtopics

- Arrays vs collections
- `List<T>`
- `Dictionary<TKey,TValue>`
- `HashSet<T>`
- `Queue<T>`
- `Stack<T>`
- `LinkedList<T>`
- `SortedList`
- `SortedDictionary`
- `SortedSet`
- Collection initialization
- Collection expressions
- `IEnumerable<T>`
- `ICollection<T>`
- `IList<T>`
- `IDictionary<TKey,TValue>`
- `IReadOnlyCollection<T>`
- `IReadOnlyList<T>`
- Read-only collections
- Mutable collections
- Immutable collections awareness

- Collection enumeration
- Collection modification
- Lookup performance
- Insertion performance
- Removal performance
- Ordering
- Uniqueness
- Choosing the correct collection
- Collection performance
- Thread-safe collection awareness

17. Generics

30 subtopics

- Generic classes
- Generic methods
- Generic interfaces
- Generic delegates
- Generic records
- Generic type parameters
- Multiple type parameters
- Generic constraints
- `where`
- `class`
- `class?`
- `struct`
- `notnull`
- `unmanaged`
- `new()`
- Interface constraints
- Base-class constraints
- Multiple constraints
- Generic type safety
- Generic reusability
- Generics and boxing
- Open generic types
- Closed generic types
- Covariance
- Contravariance
- Generic variance
- Generic DI registrations
- Generic repository concepts
- Generic result types
- Generic factory concepts

18. Delegates, Lambdas, Events and Expression Trees

31 subtopics

- Delegate
- Delegate declaration
- Delegate instances
- Method groups
- Multicast delegates
- `Action`

- `Func`
- `Predicate`
- Anonymous methods
- Lambda expressions
- Expression lambdas
- Statement lambdas
- Lambda parameters
- Lambda return values
- Closures
- Captured variables
- Closure pitfalls
- Events
- Event publishers
- Event subscribers
- `EventHandler`
- `EventArgs`
- Custom events
- Event subscription
- Event unsubscription
- Memory leak awareness
- Observer relationship
- Expression trees
- `Expression<TDelegate>`
- Compiled expressions
- Expression trees in LINQ providers

19. LINQ

54 subtopics

- LINQ fundamentals
- Method syntax
- Query syntax
- `IEnumerable<T>`
- `IQueryable<T>`
- `Where`
- `Select`
- `SelectMany`
- `OrderBy`
- `OrderByDescending`
- `ThenBy`
- `ThenByDescending`
- `GroupBy`
- `Join`
- `GroupJoin`
- `Any`
- `All`
- `Contains`
- `First`
- `FirstOrDefault`
- `Single`
- `SingleOrDefault`
- `Last`

- `LastOrDefault`
- `Count`
- `LongCount`
- `Sum`
- `Average`
- `Min`
- `Max`
- `Aggregate`
- `Distinct`
- `DistinctBy`
- `Union`
- `Intersect`
- `Except`
- `Skip`
- `Take`
- `Chunk`
- `Cast`
- `OfType`
- `ToList`
- `ToArray`
- `ToDictionary`
- `ToHashSet`
- Deferred execution
- Immediate execution
- Multiple enumeration
- Projection
- Query composition
- LINQ performance
- LINQ to Objects
- LINQ to Entities
- Client vs server execution awareness

20. Exception Handling and Resource Management

30 subtopics

- Exception hierarchy
- Built-in exceptions
- `try`
- `catch`
- `finally`
- `throw`
- `throw;`
- Multiple catch blocks
- Exception filters
- Inner exceptions
- Custom exceptions
- Exception propagation
- Stack traces
- Exception wrapping
- Exception logging
- Exceptions vs validation
- Exceptions vs business failures

- Exceptions vs Result pattern
- `IDisposable`
- Dispose
- Dispose pattern
- `using`
- Using declarations
- `IAsyncDisposable`
- `await using`
- Resource ownership
- Resource cleanup
- Finalizer awareness
- Safe exception handling
- Avoiding swallowed exceptions

21. Files, Streams and Serialization

31 subtopics

- File operations
- Directory operations
- `File`
- `Directory`
- `FileInfo`
- `DirectoryInfo`
- Streams
- `Stream`
- `FileStream`
- `MemoryStream`
- `BufferedStream`
- `StreamReader`
- `StreamWriter`
- Binary streams awareness
- Async file operations
- File encoding
- JSON
- `System.Text.Json`
- Serialization
- Deserialization
- JSON options
- Naming policies
- Null handling
- Enum serialization
- Date serialization
- Custom converters
- Reference cycles
- Ignore attributes
- Large JSON handling
- Streaming JSON awareness
- XML serialization awareness

22. .NET Runtime, Reflection and Memory

39 subtopics

- CLR
- Managed code

- IL
- JIT
- Assemblies
- Metadata
- Assembly loading
- Type system
- Reflection
- `Type`
- `GetType`
- `typeof`
- `PropertyInfo`
- `MethodInfo`
- `ConstructorInfo`
- Attributes
- Custom attributes
- Runtime type inspection
- Dynamic object creation
- Dynamic invocation
- Reflection performance
- Garbage collection
- GC generations
- Gen 0
- Gen 1
- Gen 2
- Large Object Heap
- Object allocation
- Finalizers
- GC pressure
- `Span<T>`
- `ReadOnlySpan<T>`
- `Memory<T>`
- `ArrayPool<T>`
- Allocation reduction
- Profiling
- Benchmarking awareness
- Native AOT awareness
- Trimming awareness

23. Async Programming

38 subtopics

- Synchronous execution
- Asynchronous execution
- `Task`
- `Task<T>`
- `async`
- `await`
- Async method return types
- Async exception handling
- Sequential awaits
- Concurrent awaits
- `Task.WhenAll`

- `Task.WhenAny`
- `Task.Delay`
- CancellationToken
- CancellationTokenSource
- Cancellation propagation
- Cooperative cancellation
- Timeouts
- I/O-bound work
- CPU-bound work
- `Task.Run`
- Async streams
- `IAsyncEnumerable<T>`
- `await foreach`
- `ValueTask`
- `IAsyncDisposable`
- Async disposal
- `.Result`
- `.Wait()`
- Sync-over-async
- Fire-and-forget risks
- Async method naming
- Async all the way
- Async in controllers
- Async in EF Core
- Async HttpClient calls
- Cancellation in APIs
- Async performance considerations

24. Threading, Synchronization and Parallel Programming

38 subtopics

- Process vs thread
- Thread
- ThreadPool
- Task scheduler
- Shared state
- Race conditions
- Thread safety
- Deadlocks
- Starvation
- Atomic operations
- `lock`
- `Monitor`
- `Interlocked`
- `Semaphore`
- `SemaphoreSlim`
- Mutex
- ReaderWriterLockSlim
- SynchronizationContext
- ConcurrentDictionary
- ConcurrentQueue
- ConcurrentStack

- ConcurrentBag
- Channels
- Producer-consumer
- Parallel programming
- `Parallel.For`
- `Parallel.ForEach`
- `Parallel.ForEachAsync`
- PLINQ
- Degree of parallelism
- Parallel cancellation
- Parallel exceptions
- CPU-bound parallelism
- Async vs threading
- Async vs parallelism
- Thread-pool starvation
- Parallelism in ASP.NET Core
- Avoiding uncontrolled parallel work

25. Clean Code, SOLID and Refactoring

35 subtopics

- Clean code principles
- Naming
- Method design
- Class design
- Single responsibility
- Separation of concerns
- DRY
- KISS
- YAGNI
- Guard clauses
- Immutability
- Side-effect control
- Code smells
- Long methods
- Large classes
- Primitive obsession
- Duplicate code
- Shotgun surgery awareness
- Feature envy awareness
- Refactoring
- Extract method
- Extract class
- Rename refactoring
- SOLID
- Single Responsibility Principle
- Open/Closed Principle
- Liskov Substitution Principle
- Interface Segregation Principle
- Dependency Inversion Principle
- Coupling
- Cohesion

- Loose coupling
- High cohesion
- Testable code
- Avoiding over-engineering

26. Design Patterns

54 subtopics

- Pattern purpose
- Pattern selection
- Creational patterns
- Factory Pattern
- Simple Factory
- Factory Method
- Abstract Factory awareness
- Builder Pattern
- Host Builder
- WebApplicationBuilder
- Singleton Pattern
- Singleton DI lifetime distinction
- DbContext Factory
- Structural patterns
- Repository Pattern
- Generic Repository
- Unit of Work Pattern
- Decorator Pattern
- `DelegatingHandler`
- Adapter Pattern
- Facade Pattern
- Proxy Pattern
- Lazy initialization
- Composite Pattern awareness
- Behavioral patterns
- Strategy Pattern
- Template Method
- Chain of Responsibility
- Observer Pattern
- Command Pattern
- Mediator Pattern
- State Pattern
- Specification Pattern
- Result Pattern
- Dependency Injection Pattern
- Service Layer Pattern
- DTO Pattern
- Mapper Pattern
- AutoMapper
- Options Pattern
- Middleware Pattern
- MVC Pattern
- Policy Scheme Pattern
- Exception Handler Pattern

- CQRS Pattern
- Domain Event Pattern
- Integration Event Pattern
- Outbox Pattern
- Retry Pattern
- Circuit Breaker Pattern
- Cache-Aside Pattern
- Idempotency Pattern
- When not to use patterns
- Pattern over-engineering

27. Dependency Injection and Service Lifetimes

34 subtopics

- Inversion of Control
- Dependency Injection
- Constructor injection
- Method injection awareness
- Property injection awareness
- `IServiceCollection`
- `IServiceProvider`
- Service registration
- Service resolution
- Transient
- Scoped
- Singleton
- Lifetime selection
- Scoped-per-request
- Captive dependency
- Singleton consuming scoped services
- Thread-safe singleton services
- Factory registration
- Delegate registration
- Open generic registrations
- Multiple implementations
- `IEnumerable<T>` injection
- Keyed services
- `IServiceScope`
- `IServiceScopeFactory`
- Creating manual scopes
- Service disposal
- Options injection
- DI extension methods
- Registration modules
- Service Locator anti-pattern
- DI and testing
- DI validation
- DI best practices

28. HTTP, HTTPS and REST

50 subtopics

- HTTP protocol
- HTTP request

- HTTP response
- HTTP versions awareness
- HTTPS
- TLS awareness
- URL
- URI
- Scheme
- Host
- Port
- Path
- Query string
- Headers
- Request body
- Response body
- Content type
- Media types
- JSON
- Cookies
- Authorization header
- GET
- POST
- PUT
- PATCH
- DELETE
- HEAD
- OPTIONS
- Safe methods
- Idempotent methods
- HTTP status codes
- 2xx
- 3xx
- 4xx
- 5xx
- REST
- Resources
- Representations
- Resource naming
- Statelessness
- API contracts
- Idempotency
- Pagination
- Filtering
- Sorting
- Searching
- Error contracts
- Content negotiation awareness
- HTTP caching awareness
- ETags awareness

29. ASP.NET Core Fundamentals

30 subtopics

- ASP.NET Core architecture
- Generic Host
- Hosting model
- Kestrel
- `Program.cs`
- `WebApplicationBuilder`
- `WebApplication`
- Minimal hosting model
- Application startup
- Service registration
- Application build
- Application run
- Application lifetime
- Request pipeline
- Endpoints
- Endpoint routing
- Environment
- Development
- Staging
- Production
- Configuration
- Logging
- Dependency injection
- Host configuration
- App configuration
- Graceful shutdown
- HTTPS redirection
- Developer exception page
- Static files awareness
- Reverse proxy awareness

30. Configuration, Options and Secrets

27 subtopics

- `appsettings.json`
- Environment-specific appsettings
- Environment variables
- Command-line configuration
- Configuration providers
- Configuration precedence
- `IConfiguration`
- Configuration sections
- Strongly typed configuration
- Options Pattern
- `IOptions<T>`
- `IOptionsSnapshot<T>`
- `IOptionsMonitor<T>`
- Named options
- Options validation
- Validate on start

- Configuration reload
- User Secrets
- Connection strings
- API keys
- Secrets
- Secret rotation awareness
- Environment-specific secrets
- Azure Key Vault
- Managed Identity awareness
- Configuration best practices
- Avoiding secrets in source control

31. Middleware, Routing, Controllers and Minimal APIs

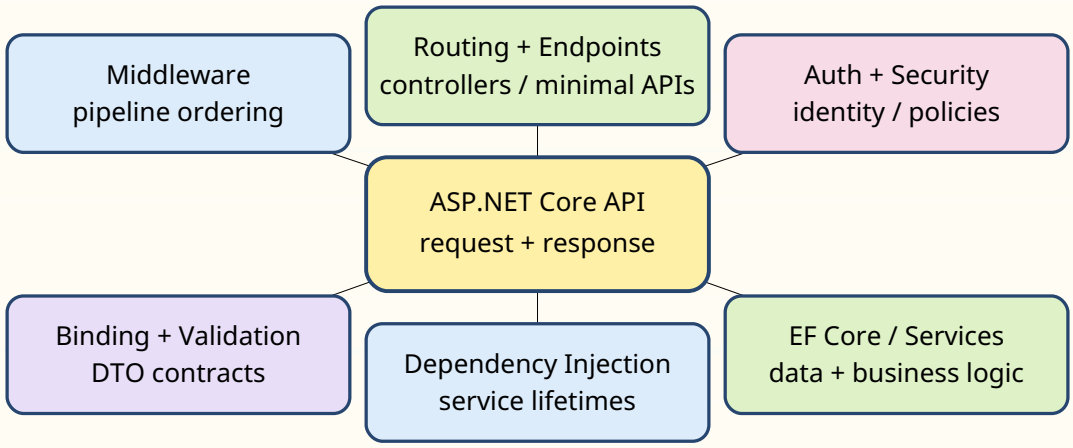
45 subtopics

- Middleware
- Request pipeline
- Middleware ordering
- `Use`
- `Run`
- `Map`
- Short-circuiting
- Built-in middleware
- Custom middleware
- Exception middleware
- Authentication middleware
- Authorization middleware
- CORS middleware
- Routing
- Endpoint routing
- Attribute routing
- Route templates
- Route parameters
- Optional route parameters
- Route constraints
- Query parameters
- Named routes
- Controllers
- `ControllerBase`
- `[ApiController]`
- `[Route]`
- `[HttpGet]`
- `[HttpPost]`
- `[HttpPut]`
- `[HttpPatch]`
- `[HttpDelete]`
- Action methods
- `IActionResult`
- `ActionResult<T>`
- Controller DI
- Minimal APIs
- `MapGet`

- `MapPost`
- `MapPut`
- `MapPatch`
- `MapDelete`
- Route groups
- Endpoint filters
- Typed results
- Controllers vs Minimal APIs

REFERENCE DIAGRAM 2

ASP.NET Core Web API Building Blocks



32. Model Binding, DTOs, Mapping and Validation

37 subtopics

- Model binding
- Binding sources
- `[FromRoute]`
- `[FromQuery]`
- `[FromBody]`
- `[FromHeader]`
- `[FromForm]`
- Complex binding
- Collection binding
- DTO Pattern
- Request DTOs
- Response DTOs
- Create DTOs
- Update DTOs
- Patch models awareness
- Summary DTOs
- Detail DTOs
- Entity vs DTO
- API contracts
- Manual mapping
- Mapper Pattern
- AutoMapper
- AutoMapper profiles
- Projection mapping

- Mapping validation
- Over-posting prevention
- Model validation
- Data annotations
- Required validation
- Length validation
- Range validation
- Regex validation
- Custom validation
- Cross-property validation
- Business-rule validation
- Async validation
- Validation error responses

33. API Responses, Errors, Filters, OpenAPI and Versioning

46 subtopics

- `Ok`
- `Created`
- `CreatedAtAction`
- `NoContent`
- `BadRequest`
- `NotFound`
- `Conflict`
- `Unauthorized`
- `Forbid`
- Status code responses
- Response headers
- Location header
- Consistent response contracts
- Global exception handling
- `IExceptionHandler`
- Exception Handler Pattern
- ProblemDetails
- ValidationProblemDetails
- Exception-to-status-code mapping
- Business exceptions
- Domain exceptions
- Safe error messages
- Trace IDs
- Error logging
- Authorization filters
- Resource filters
- Action filters
- Exception filters
- Result filters
- Filter ordering
- Custom filters
- Middleware vs filters
- OpenAPI
- API schemas
- Endpoint descriptions

- Response documentation
- Authentication documentation
- API versioning
- URL versioning
- Query-string versioning
- Header versioning
- Media-type versioning
- Default API version
- Version reporting
- API deprecation
- Backward compatibility

34. EF Core Fundamentals and Modeling

47 subtopics

- EF Core architecture
- SQL Server provider
- `DbContext`
- `DbSet<T>`
- DbContext lifetime
- `IDbContextFactory<TContext>`
- Connection strings
- Entity classes
- Primary keys
- Composite keys
- Foreign keys
- Alternate keys awareness
- Required properties
- Optional properties
- Data annotations
- Fluent API
- EntityTypeConfiguration
- One-to-one
- One-to-many
- Many-to-many
- Navigation properties
- Join entities
- Required relationships
- Optional relationships
- Delete behavior
- Cascade delete
- Restrict behavior
- Indexes
- Unique indexes
- Decimal precision
- String length
- Value converters
- Owned types
- Complex types awareness
- Shadow properties
- Migrations
- Migration history

- Migration scripts
- Seed data
- Change tracking
- Entity states
- Add
- Update
- Remove
- Attach
- SaveChanges
- SaveChangesAsync

35. EF Core Queries, Transactions, Concurrency and Performance

49 subtopics

- LINQ to Entities
- `IQueryable<T>`
- Query translation
- Projection
- `Select`
- `Where`
- `Include`
- `ThenInclude`
- `AsNoTracking`
- Tracking queries
- No-tracking queries
- Identity resolution awareness
- Eager loading
- Explicit loading
- Lazy loading awareness
- N+1 problem
- Split queries
- Single queries
- Filtering
- Sorting
- Pagination
- Aggregations
- Generated SQL
- Query tags awareness
- Raw SQL
- Parameterized SQL
- Stored procedures
- Transactions
- Begin transaction
- Commit
- Rollback
- Transaction boundaries
- Savepoints awareness
- Optimistic concurrency
- Concurrency tokens
- RowVersion
- `DbUpdateConcurrencyException`
- Retry handling

- Global query filters
- Interceptors
- Execution strategies
- Connection resiliency
- Query performance
- Index awareness
- Compiled queries awareness
- Bulk operations awareness
- Repository with EF Core
- Unit of Work with EF Core
- When not to wrap EF Core unnecessarily

36. Authentication, Authorization and Security

57 subtopics

- Authentication
- Authorization
- Authentication schemes
- Authentication handlers
- ClaimsPrincipal
- ClaimsIdentity
- Claims
- Roles
- JWT
- JWT header
- JWT payload
- JWT signature
- Access tokens
- Refresh tokens
- Issuer
- Audience
- Expiration
- Signing keys
- Token validation
- Bearer authentication
- ASP.NET Core Identity
- Users
- Roles
- Password hashing
- UserManager
- SignInManager
- RoleManager
- Claims-based authorization
- Role-based authorization
- Policy-based authorization
- Authorization requirements
- Authorization handlers
- Resource-based authorization
- Tenant authorization
- Ownership authorization
- OAuth 2.0
- OpenID Connect

- Authorization Code Flow
- PKCE
- Microsoft Entra ID awareness
- Policy Scheme Pattern
- Scheme forwarding
- CORS
- Same-origin policy
- HTTPS
- CSRF awareness
- XSS awareness
- SQL injection prevention
- Over-posting prevention
- Input validation
- Secure headers
- Rate limiting
- Request size limits
- Secret management
- Sensitive data logging
- Least privilege
- Dependency vulnerability awareness

37. xUnit and .NET Testing

67 subtopics

- Testing fundamentals
- Unit tests
- Integration tests
- Functional tests
- End-to-end tests
- xUnit
- `[Fact]`
- `[Theory]`
- `[InlineData]`
- `[MemberData]`
- `[ClassData]`
- Assertions
- Arrange-Act-Assert
- Test naming conventions
- Positive tests
- Negative tests
- Boundary tests
- Async tests
- Exception tests
- Test isolation
- Test fixtures
- `IClassFixture<T>`
- Collection fixtures
- `IAsyncLifetime`
- Shared test context
- Test data builders
- Object Mother awareness
- Mocking

- Stubs
- Fakes
- Spies awareness
- Test doubles
- Moq
- NSubstitute awareness
- Mock verification
- Mock behavior
- Avoiding over-mocking
- Testing services
- Testing repositories
- Testing Unit of Work
- Testing factories
- Testing strategies
- Testing decorators
- Testing exception handlers
- Testing middleware
- Controller testing
- Minimal API testing
- `WebApplicationFactory`
- `TestServer`
- HttpClient integration testing
- Authentication testing
- Authorization testing
- Policy testing
- EF Core testing
- SQL Server integration testing
- Test database management
- Database cleanup
- Test transactions
- InMemory provider limitations
- SQLite testing awareness
- Contract testing awareness
- Snapshot testing awareness
- Code coverage
- Mutation testing awareness
- Performance testing
- Load testing
- Smoke testing

38. Logging, Observability, Health and Diagnostics

36 subtopics

- `ILogger<T>`
- Log levels
- Trace
- Debug
- Information
- Warning
- Error
- Critical
- Structured logging

- Message templates
- Logging scopes
- Correlation IDs
- Trace IDs
- Request logging
- Exception logging
- Sensitive-data redaction awareness
- Metrics
- Counters awareness
- Traces
- Distributed tracing
- OpenTelemetry
- Activities
- Trace context
- Health checks
- Liveness
- Readiness
- Database health checks
- External service health checks
- Custom health checks
- Application Insights
- Azure Monitor awareness
- Production diagnostics
- Monitoring
- Alerts
- Dashboards awareness
- Log aggregation awareness

39. Caching, HttpClient, Resilience and Background Processing

42 subtopics

- In-memory caching
- Distributed caching
- Output caching
- Response caching
- Cache keys
- Cache expiration
- Absolute expiration
- Sliding expiration
- Cache invalidation
- Cache stampede awareness
- Tenant-aware caching
- Cache-Aside Pattern
- HttpClient
- `IHttpClientFactory`
- Named clients
- Typed clients
- Base addresses
- Default headers
- JSON HTTP calls
- Timeouts
- Cancellation

- `DelegatingHandler`
- Decorator with HttpClient
- Retry
- Exponential backoff
- Jitter
- Circuit breaker
- Resilience pipelines
- Transient errors
- Idempotency
- `IHostedService`
- `BackgroundService`
- Queued background work
- Channels
- Producer-consumer
- Scoped dependencies in workers
- Graceful shutdown
- Background retries
- Duplicate handling
- Idempotent processing
- Long-running work
- Scheduled jobs awareness

40. Architecture, Production, Docker, Azure and Angular Integration

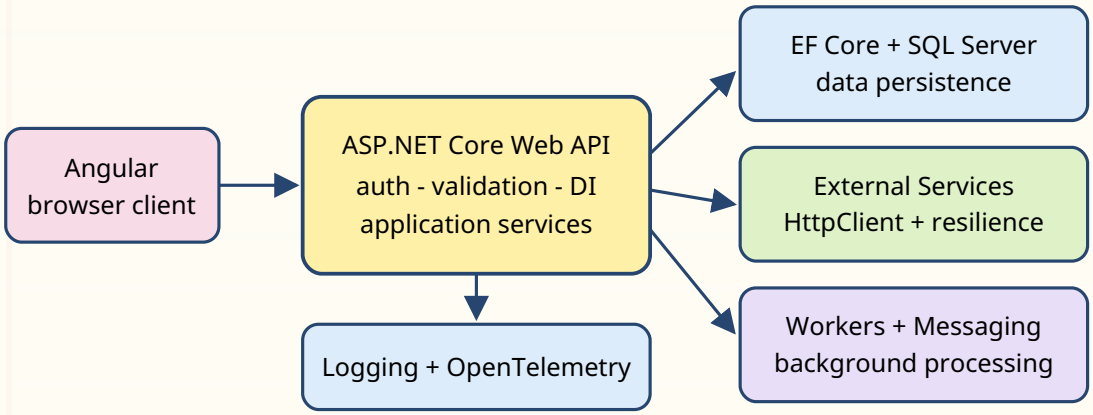
72 subtopics

- Layered Architecture
- Clean Architecture
- Vertical Slice Architecture
- Modular Monolith
- Feature folders
- Domain layer
- Application layer
- Infrastructure layer
- API layer
- Dependency direction
- Domain-Driven Design fundamentals
- Entities
- Value objects
- Aggregates
- Aggregate roots
- Domain services
- Application services
- Domain events
- Integration events
- CQRS
- Commands
- Queries
- Command handlers
- Query handlers
- Mediator pattern
- Outbox Pattern
- Eventual consistency

- Idempotency keys
- File uploads
- File downloads
- Streaming
- Response compression
- API performance
- Profiling
- Load testing
- Docker
- Dockerfile
- Multi-stage builds
- Container images
- Containers
- Environment variables
- Container networking
- Docker Compose awareness
- `dotnet publish`
- Framework-dependent deployment
- Self-contained deployment
- Reverse proxy
- Production configuration
- Production security
- Production logging
- Production monitoring
- Azure App Service
- Azure Container Apps
- Azure SQL
- Azure Key Vault
- Managed Identity
- Azure Storage
- Azure Service Bus
- Application Insights
- Azure Monitor
- Angular API base URL
- Angular HttpClient integration
- CORS with Angular
- JWT with Angular
- Refresh-token integration

REFERENCE DIAGRAM 3

.NET Web API Production Integration Landscape



- ProblemDetails integration
- Pagination contracts
- Filtering contracts
- Sorting contracts
- File upload contracts
- Date/time contracts
- Deployment configuration

Roadmap rule: learn the topics in order, but revisit earlier C# concepts while building ASP.NET Core, EF Core, testing, security, and production features.